



Book I. Prop. XXIX. Theor. *A*

straight line (—) falling on two parallel straight lines (— and —), makes the alternate angles equal to one another; and also the external equal to the internal and opposite angle on the same side; and the two internal angles on the same side together equal to two right angles. For

if the alternate angles  and  be not equal, draw —, making  =  (pr. I.23).

Therefore — — — — || — (pr. I.27) and therefore two straight lines which intersect are parallel to the same straight line, which is impossible (ax. XII).

Hence  and  are not unequal, that is, they are equal:  =  (pr. I.15); $\therefore$  = , the external angle equal to the internal and opposite on the same side: if  be added to both, then  +  =

 =  (pr. I.13). That is to

say, the two internal angles at the same side of the cutting line are equal to two right angles.

Q. E. D.